

Measuring the Defenders: A Layer-Aware, Framework-Mapped Benchmark for Model Context Protocol Security Proxies

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Abstract

The Model Context Protocol (MCP) has become a common way to connect large-language-model (LLM) agents to external tools and data, and a substantial 2025–2026 literature now documents its attack surface. Existing security benchmarks for MCP measure how well an **LLM agent resists** attacks. None, to our knowledge, measure how well a **defensive proxy or gateway covers** that attack surface, and none map their results to the governance frameworks organizations actually use — the NIST AI Risk Management Framework (AI RMF) and the OWASP Top 10 for LLM and Agentic Applications. We present two artifacts to fill this gap: (1) a **threat–control crosswalk** of 22 MCP attack vectors mapped to architectural layer, STRIDE class, NIST AI RMF function, and OWASP category; and (2) an open, vendor-neutral **defense-side benchmark** that runs a candidate MCP security proxy against a corpus of attack fixtures with matched benign controls, and scores its coverage using the crosswalk as a rubric. Each tool is driven through its *real* code, not a mock. We evaluate three open-source proxies (mcp-bastion, mcp-firewall, pipelock) plus a null baseline over 28 test cases. The strongest proxy reaches **34% weighted coverage (13 of 22 vectors)** — a proxy’s realistic ceiling — and even the union of all three tools leaves **9 of 22 vectors covered by none**: the entire registry/supply-chain surface, OS-isolation-dependent vectors, and several semantic vectors. All results carry zero false positives against matched benign controls. We argue this is direct, quantified evidence that a runtime proxy alone is insufficient and that defense-in-depth spanning *distinct mechanism classes* (proxy, cryptographic attestation, OS isolation) is required. We also report two methodological findings we observed first-hand: benchmark scores are sensitive to how attack fixtures are encoded (a fairness hazard), and the benchmark can *guide* a defender’s development — over the study, mcp-bastion’s coverage rose from 9% to 34% as gaps it surfaced were fixed and re-verified. A matched benign-control design and a “measure only what the tool inspects at runtime” integrity rule keep the results honest.

1. Introduction

MCP standardizes how an LLM agent discovers and calls external “tools” exposed by “servers.” Its rapid adoption across consumer and enterprise agents has been accompanied, in 2025–2026, by a dense body of security research: systematization-of-knowledge papers, formal threat taxonomies,

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attack benchmarks, and cryptographic attestation proposals (Section 2). This work establishes *what can go wrong*.

A practical question remains under-addressed: **given a defensive tool that sits in front of MCP servers — a proxy, gateway, or scanner — how much of the known attack surface does it actually cover?** Answering it well requires three things that, individually, exist in the literature but have not been combined:

1. a **shared taxonomy** of MCP attack vectors;
2. a mapping from those vectors to the **governance frameworks** (NIST AI RMF, OWASP) that security and procurement teams reason in; and
3. an **executable, reproducible** way to test a defender and report coverage.

We contribute all three, plus an empirical study. Concretely:

- **A threat-control crosswalk** (Section 3): 22 MCP attack vectors, each mapped to one or more of six architectural layers, a STRIDE class, NIST AI RMF functions, and OWASP LLM-2025 and Agentic-2026 categories. Published as open, versioned, machine-readable data.
- **A defense-side benchmark** (Section 4): for each vector, one or more attack **fixtures** paired with a near-identical **benign control**; a small **adapter** per tool that drives the tool’s real detection code; and a scorer that reports coverage per layer and per framework.
- **An empirical evaluation** (Sections 5–6) of three open-source proxies and a null baseline, yielding the complementarity finding above.
- **A candid methodology discussion** (Sections 7–8): the corpus-encoding sensitivity we observed, the matched-control and runtime-integrity safeguards, and the limits of the study.

Our aim is not to rank products but to map coverage, expose gaps, and provide reusable infrastructure.

2. Related Work

All 2603/2604-series works below are recent, non-peer-reviewed preprints whose taxonomies are their authors’ own constructs; we cite them as *proposed by*, not as established consensus. Coverage statistics quoted from any single paper reflect that paper’s surveyed set.

Threat taxonomies. Recent work systematizes MCP threats through several lenses: an SoK separating adversarial security threats (indirect prompt injection, tool poisoning) from epistemic safety hazards across the Resources/Prompts/Tools primitives [SoK-2512.08290]; a formal framework of 7 threat categories and 23 attack vectors grounded in a large tool corpus [Formal-2604.05969]; a STRIDE/DREAD model over five MCP components that identifies tool poisoning as the most impactful client-side vulnerability [STRIDE-2603.22489]; and a defense-placement taxonomy classifying defenses by the layer that enforces them, which finds protection “uneven and predominantly tool-centric” with weak transport, host-orchestration, and supply-chain coverage [MCP-DPT-2604.07551]. Our crosswalk reuses these vector families but adds the framework mapping none of them provides.

Attack-side benchmarks. MSB [MSB-2510.15994] (12 attacks, ~2,000 instances, real MCP execution) measures the resistance of the **LLM agent** across the tool-use pipeline; MCPTox [MCPTox-2508.14925] scopes to tool poisoning; AgentDefense-Bench aggregates datasets into MCP-JSON-RPC test cases; MCP-Bench [MCPBench-2508.20453] measures agent *capability*, not security. All

measure the agent (or the model), not a **defensive proxy**. Ours measures the defender, on the full mapped surface, and reports coverage crosswalked to NIST/OWASP — which we could not find in any surveyed benchmark.

Attestation and provenance. ETDI [ETDI-2506.01333] proposes signed, versioned tool definitions; a “Trustworthy MCP Registry” blueprint [TrustReg-FI2026] composes existing standards (well-known URIs, Sigstore, JSON Canonicalization + JWS) for provenance and runtime integrity, and notes the official MCP registry remains an unverified pointer architecture. These are complementary to our benchmark; a future attestation-conformance suite (Section 9) would test them.

Scanners and gateways. Practitioner tools include Invariant Labs’ MCP-Scan (since acquired by Snyk and now a cloud-gated scanner), Lasso’s and EnkryptAI’s MCP gateways, Docker’s isolation-based MCP gateway, and the proxies we evaluate here. Our benchmark scores locally-deterministic defenders and treats cloud-model defenders as observable-but-not-reproducibly-scoreable (Section 4.4).

3. The Threat–Control Crosswalk

The crosswalk is the benchmark’s rubric and a standalone artifact. It enumerates **22 MCP attack vectors** consolidated from the taxonomies in Section 2, and maps each to:

- **Architectural layer** (one of: host-orchestration, client, transport, server, tool, registry-supply-chain) — following the defense-placement view of [MCP-DPT-2604.07551];
- **STRIDE** class;
- **NIST AI RMF** function(s): Govern, Map, Measure, Manage;
- **OWASP Top 10 for LLM Applications (2025)** category (LLM01–LLM10); and
- **OWASP Top 10 for Agentic Applications (2026)** category (ASI01–ASI10).

Table 1 lists the vectors and their primary layer. The full mapping, including STRIDE and all framework codes, is published as machine-readable data (`rubric/crosswalk.json`, CC-BY-4.0).

Table 1. The 22 vectors (primary layer).

#	Vector	Layer	#	Vector	Layer
1	Tool poisoning	tool	12	Configuration drift	server
2	Tool shadowing / name collision	client	13	Sandbox escape	server
3	Rug pull (definition mutation)	tool	14	Schema / validation bypass	server
4	Out-of-scope parameter injection	tool	15	Man-in-the-middle (transport)	transport
5	Prompt injection via tool results	tool	16	DNS rebinding (local servers)	transport

#	Vector	Layer	#	Vector	Layer
6	Indirect / retrieval injection	tool	17	Server impersonation	registry- supply-chain
7	Cross-tool exfiltration (confused deputy)	client	18	Excessive permission / escalation	host- orchestration
8	Tool-transfer / cross-server chaining	host- orchestration	19	Credential / token theft	host- orchestration
9	False-error escalation	tool	20	Consent fatigue / over-broad grants	client
10	Package / name squatting	registry- supply-chain	21	Command injection	server
11	Supply-chain poisoning (provenance gap)	registry- supply-chain	22	System- prompt / context leakage	client

The mappings are indicative and reviewable, not a certification — we invite community correction, which is itself a goal of publishing the rubric openly. The OWASP Agentic categories used are the official 2026 set: ASI01 Agent Goal Hijack, ASI02 Tool Misuse and Exploitation, ASI03 Identity and Privilege Abuse, ASI04 Agentic Supply Chain Vulnerabilities, ASI05 Unexpected Code Execution (RCE), ASI06 Memory & Context Poisoning, ASI07 Insecure Inter-Agent Communication, ASI08 Cascading Failures, ASI09 Human-Agent Trust Exploitation, ASI10 Rogue Agents.

4. Benchmark Methodology

4.1 Design

For each vector the corpus contains one or more **test cases**. A test case is a static JSON object with: a **malicious fixture** (the attack, as data — a tool definition, tool call, tool result, registry entry, or scenario), a near-identical **benign control**, the **expected** defender behavior (**detect/enforce**), and a **provenance** note. Fixtures are static data, never live exploits; attacker hosts use reserved names (**attacker.example**; the cloud-metadata address **169.254.169.254** appears only as an SSRF *target* in data, never contacted).

4.2 Adapters and the runtime-integrity rule

Each evaluated tool provides a thin **adapter** exposing `assess(input) → {detect, enforce}`. The runner calls it once on the malicious fixture and once on the benign control. Adapters drive the tool’s **real** code (e.g., importing the proxy’s detection functions, or invoking its CLI/SDK with a committed configuration).

A central rule keeps scores honest:

Report only what the tool actually inspects at runtime — never what its rules *could* match if pointed at data the tool never sees.

For example, a proxy that inspects tool *definitions* but not tool *results* returns “not detected” for result-borne attacks, even if its text rules would match the result string in isolation. This measures deployed behavior, not theoretical capability.

4.3 Scoring

Per test case: if the malicious fixture is not flagged, the vector scores **none**; if it is flagged but the benign control is *also* flagged, the vector scores **none** and is marked a **false positive**; if the malicious fixture is flagged and the benign control is clean, the vector scores **detect** (warn) or **enforce** (block). A vector’s level is the best across its test cases. Weighted coverage assigns enforce = 1.0, detect = 0.5, none = 0, over 22 vectors. The matched benign control is essential: without it, a tool that flags everything would score perfectly while being useless.

4.4 Reproducibility rule

A defender is **scored** only if its detection logic runs **locally and deterministically**. Tools that delegate detection to a proprietary cloud model can be *observed* but not reproducibly *scored*, since their verdicts depend on a black box that can change between runs. This excludes, e.g., the cloud-gated Snyk agent-scan; it includes the three proxies below.

5. Experimental Setup

We evaluate four adapters over a 28-case corpus (22 base cases + 6 realistic “v2” cases added for fairness; see Section 7):

Tool	Class	License	How driven
null-baseline	control	—	returns “not detected” for all inputs
mcp-bastion	runtime proxy	Apache-2.0	imports real detection: <code>scanTool</code> , <code>scanText</code> , <code>validateArguments</code> , <code>checkRequestedScopes</code> , <code>checkTransportSecurity</code> , <code>checkRequestOrigin</code> , <code>hashToolDefinition</code>
mcp-firewall	runtime proxy (Python)	AGPL-3.0	real SDK (<code>Gateway.check / scan_response</code>) with a committed config
pipelock	egress firewall (Go)	Apache-2.0 core	real scanner via <code>explain --json</code> (offline, deterministic)

For AGPL/other licensed tools we *run* the tool to benchmark it and report scores; we do not redistribute their code. Each tool is pinned to a released version and configured with its own recommended/starter policy, committed to the repository for reproducibility.

6. Results

Table 2. Verified coverage (28 cases; weighted over 22 vectors).

Tool	Class	Weighted coverage	enforce	detect	none	False positives
mcp-bastion	runtime	34%	2	11	9	0 / 28
	proxy	(7.5/22)				
mcp-firewall	runtime	14%	3	0	19	0 / 28
	proxy	(3.0/22)				
pipelock	egress	5%	1	0	21	0 / 28
	firewall	(1.0/22)				
null-baseline	control	0%	0	0	22	0 / 28

A proxy’s ceiling is partial. mcp-bastion, the broadest tool, spans the definition, response, argument, transport, and egress/least-privilege layers (poisoning, shadowing, rug-pull; response and retrieval injection and result-borne leakage; schema/parameter validation; plaintext-transport and DNS-rebinding defense; sensitive-argument and over-broad-scope detection). mcp-firewall and pipelock add *enforcement* (deny/redact/SSRF-block) on the call and egress layers where bastion only warns.

Defense-in-depth across mechanism classes is required. Even with the strongest proxy at 34%, and across all three tools, **13 of 22 vectors are covered by at least one tool; 9 are covered by none** — the registry/supply-chain surface (package squatting, provenance-gap supply-chain poisoning, server impersonation), OS-isolation-dependent vectors (sandbox escape), and semantic vectors (tool-transfer, false-error escalation, configuration drift, command injection, consent fatigue). These are not addressable by *any* runtime proxy; they require different mechanisms — cryptographic attestation, OS sandboxing, and client-side controls. A proxy is necessary but not sufficient.

Zero false positives. No tool flagged any benign control, across all 28 cases and every feature iteration.

7. Discussion

Benchmark-guided improvement. The benchmark initially scored mcp-bastion at 9%, covering only the tool-definition layer. It then guided four rounds of proxy-native hardening, each re-measured at zero false positives: response scanning (9%→18%: response/retrieval injection, result-borne leakage), argument/schema validation (18%→23%: parameter smuggling, validation bypass), transport hardening (23%→30%: plaintext-transport warning and a DNS-rebinding Origin check that *enforces* by rejecting cross-origin requests to a loopback listener), and sensitive-argument plus least-privilege scanning (30%→34%: cross-tool exfiltration, over-broad scopes). The benchmark did not just measure the tool; it identified concrete, layer-specific gaps and verified each fix

— and then defined the proxy’s ceiling, since the remaining 9 vectors are architecturally outside a proxy’s reach.

Corpus-encoding sensitivity (a fairness hazard). We observed that scores are sensitive to how an attack is *encoded*. An early corpus expressed exfiltration with sanitized placeholder text; against it, mcp-firewall scored 5%. Adding realistic encodings (an AWS-key-shaped secret in a tool result; an exfiltration POST to the cloud-metadata address) raised it to 14% — not by changing the tool, but by probing capabilities the first corpus never exercised. We therefore treat *tool-neutral*, *realistic*, *multi-encoding* fixtures as a fairness requirement, and we caution that **absolute totals are corpus- dependent; the per-vector matrix (which vectors a tool covers) is the more reliable read.**

Adapter faithfulness. The matched benign control also guards against adapter over-reach. In wiring pipelock (an egress firewall), an initial adapter synthesized an egress URL from an *inbound* local host, causing pipelock to “block” both the malicious and benign case — surfaced immediately as a false positive and corrected by restricting URL synthesis to outbound endpoints.

8. Limitations and Threats to Validity

- **Taxonomy is a synthesis, not a standard.** The 22-vector set consolidates several proposed taxonomies; it is one reviewable synthesis. Framework mappings are indicative, single-author, and pending a second-reviewer pass; OWASP Agentic ASI05–ASI10 labels are unconfirmed.
- **Small, seeded corpus.** Most vectors have 1–2 fixtures. Absolute coverage numbers should be read as lower bounds and are corpus-dependent (Section 7).
- **Partial tool interfaces.** pipelock’s text-injection scanner (server/eval mode) is not driven; only its URL/egress scanner is measured, so its number is a lower bound on its egress capability.
- **Configuration dependence.** Results depend on each tool’s configured policy; we commit the exact configs, but different policies would yield different numbers.
- **Scope.** We score locally-deterministic proxies; cloud-model and isolation-based tools are out of scope for scoring (Section 4.4).

9. Future Work

Expand the corpus to multiple tool-neutral fixtures per vector; drive pipelock’s content-scanning mode and add further defenders (including a fully-local mode of cloud-gated gateways, if exposed); publish a public leaderboard; complete the second-reviewer pass on framework mappings and confirm ASI labels; and build an **attestation-conformance** suite to test tool-integrity/provenance schemes (ETDI, the Trustworthy Registry composition) as a coupled bench-plus-attestation artifact.

10. Conclusion

We introduced a layer-aware, framework-mapped crosswalk of 22 MCP attack vectors and an open, vendor-neutral benchmark that measures how much of that surface a defensive proxy actually covers, driving each tool through its real code. The strongest of three open-source proxies reaches 34% coverage, and 9 of 22 vectors are undefended by any measured tool — the registry/supply-chain, OS-isolation, and semantic vectors that lie outside a proxy’s reach. The evidence argues for layered defense across distinct mechanism classes, and the benchmark demonstrably guided one proxy from

9% to 34% at zero false positives. It provides a reusable, honest instrument — with matched benign controls, a runtime-integrity rule, and a reproducibility rule — for the community to measure MCP defenders and track their progress.

Availability

- **Benchmark (mcp-defense-bench):** rubric, corpus, adapters, runner, leaderboard. Rubric/data: CC-BY-4.0; harness: Apache-2.0.
- **Reference proxy (mcp-bastion):** Apache-2.0; published on npm.
- Evaluated third-party tools are used under their own licenses (mcp-firewall: AGPL-3.0; pipe-lock: Apache-2.0 core) and are not redistributed here.
- **mcp-defense-bench:** <https://github.com/Gowthaman90/mcp-defense-bench>
- **mcp-bastion:** <https://github.com/Gowthaman90/mcp-bastion> · npm: `mcp-bastion`

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